

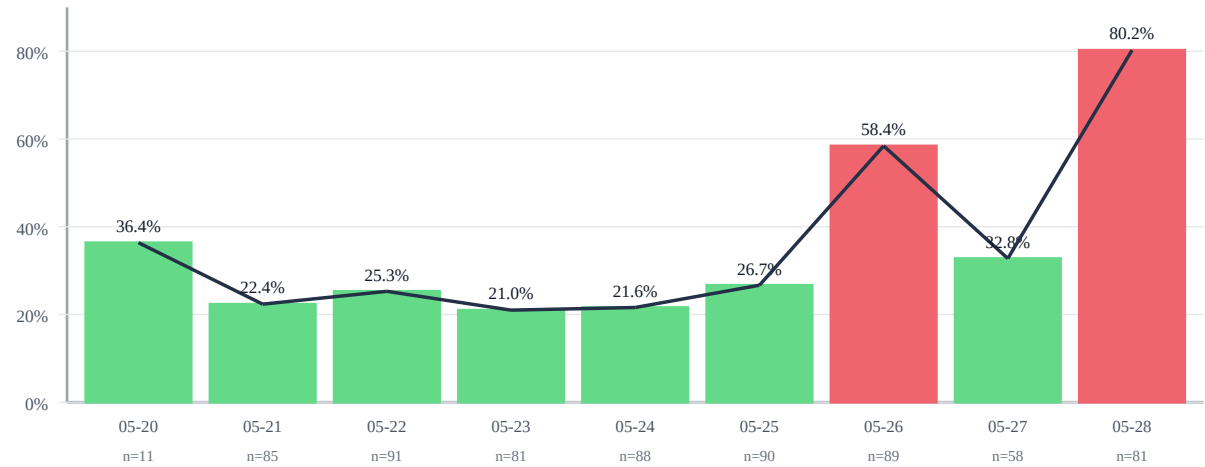
Explora 3B Reboot Investigation

Management report from supplied CSV, raw compressed decoder logs, and prior RCA PDF | Prepared 28 May 2026

Executive verdict: the reboot concern is real and material. The CSV shows a stable multi-reboot baseline of roughly 25.5% through 25 May, then a step-change to 58.4% on 26 May and 80.2% on 2026-05-28. That pattern is not consistent with 98 independent decoder failures. The most likely root cause is a shared broadcast/OTA/CAS trigger acting on a vulnerable decoder population; chronic units also show strong local signal/hardware degradation.

Metric	Result	Meaning
Fleet rows analysed	674 rows / 98 smartcards	Full dashboard export, 20-28 May 2026
Peak daily impact	80.2% multi-reboot cards on 2026-05-28	Fleet event threshold breached
Raw logs analysed	125 logs across 9 chronic smartcards	Deep evidence set for top offenders
Boot confirmation	125/125 logs open with cold boot script	Log_Count is a valid reboot proxy
Explicit software reboot path	32/125 logs	Some events are application/middleware-triggered restarts
Satellite front-end fault	124/125 logs, 265,763 total Si2183 errors	Signal path/tuner fault is pervasive in chronic units
OTA 1970 epoch scheduling	117/125 logs	Confirmed software/time defect candidate, not universal in this bundle

Share of smartcards with >1 reboot log per day



Root Cause Assessment

1. The reboots are genuine cold starts, not just standby wakes or logging artefacts.

Every raw log begins with the ``startIDwayJ.sh`` boot sequence and the CSV counts multiple boot-log files for the same smartcard on the same day. The 28 May position is especially severe: 65 of 81 active smartcards had more than one reboot log.

2. The fleet shape points to a shared trigger.

Before 26 May, the proportion of devices with multiple daily reboot logs sits near the normal/chronic range. The sharp common-mode jump on 26 May and again on 28 May points upstream: OTA signalling, CAS/entitlement changes, SI/broadcast table changes, or a footprint-wide signal event. A pure hardware-aging explanation cannot account for the synchronised rise.

3. Chronic devices have real local decoder/signal faults.

The raw logs show Si2183 satellite demodulator errors in 124 of 125 logs. Several units also show SI table CRC churn, filter timeouts, network-unavailable messages, and HDD temperature command failures. These issues can make specific decoders more likely to reboot under a broadcast or OTA stressor.

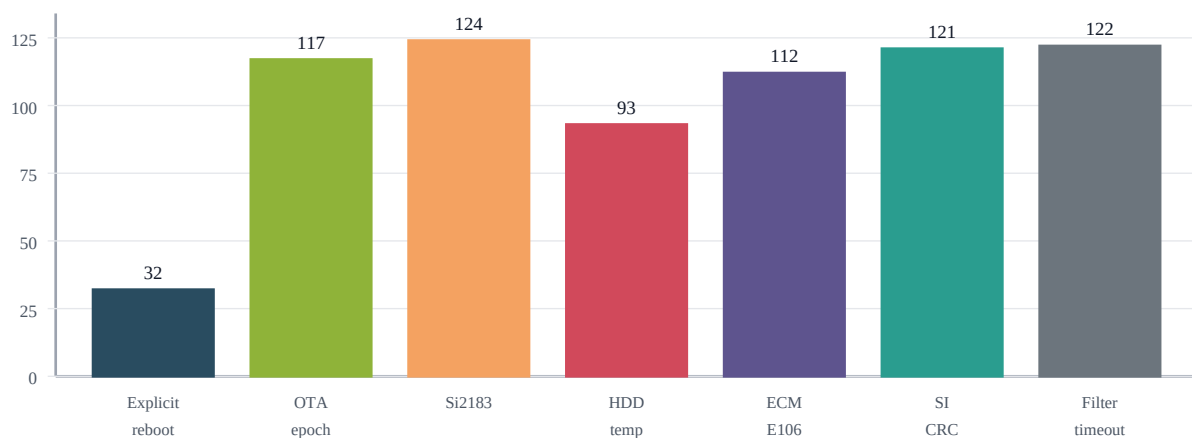
4. The strongest software lead remains boot-time scheduling before valid time is established.

The 1970 epoch OTA maintenance signature appears in 117 raw logs, and OTA v5.1 appears in 124 logs. Where present, this is a credible mechanism: a decoder that schedules maintenance before NTP/DVB time sync may treat the maintenance time as already due and re-enter update/restart logic. Because it is not present in every log in this bundle, it should be treated as a high-priority software defect candidate rather than the only cause.

5. Some restarts are explicitly requested by middleware.

32 logs include the controlled path ``Terminate(reboot=true)`` or ``Rebooting the STB``. That is important: for those cases the box is not simply losing power. The initiating subsystem must be correlated with OTA/CAS/SI events immediately before the termination.

Fault signatures across 125 raw boot logs



Evidence by Chronic Decoder

Smartcard	MAC	CSV logs	Raw logs	Avg uptime	Si2183 errors	Max BER	Explicit reboot logs	RCA note
4664264050	78:6A:1F:E5:A9:F6	42	42	1.6h	21,669	-	4	controlled reboot, signal/tuner, HDD temp, OTA epoch
4664263996	78:6A:1F:F0:48:5E	21	21	5.4h	25,833	-	0	signal/tuner, HDD temp, OTA epoch
4664263958	78:6A:1F:F0:47:BE	13	13	2.6h	7,747	-	0	signal/tuner, HDD temp, OTA epoch
4664265104	C0:89:AB:ED:3B:47	10	10	16.4h	44,492	-	6	controlled reboot, signal/tuner, HDD temp, OTA epoch
4671199079	0C:7F:B2:D8:4C:25	10	10	20.4h	54,773	-	7	controlled reboot, signal/tuner, HDD temp, OTA epoch
4664264086	78:6A:1F:F0:48:FB	8	8	19.3h	38,146	-	4	controlled reboot, signal/tuner, HDD temp, OTA epoch
4664265184	A4:98:13:55:34:D1	8	8	22.0h	39,881	-	6	controlled reboot, signal/tuner, OTA epoch
4664264683	78:6A:1F:E5:A9:68	7	7	6.4h	9,856	-	2	controlled reboot, signal/tuner, HDD temp, OTA epoch
4664263783	C0:89:AB:ED:45:2D	6	6	17.0h	23,366	-	3	controlled reboot, signal/tuner, HDD temp, OTA epoch

Recommended Management Actions

Priority	Action	Owner	Why it matters
P1	Overlay OTA push/load-sequence history, CAS re-key/EMM/ECM changes, and SI table updates for 25-28 May against reboot timestamps.	Broadcast / OTA / CAS	Confirms or clears the shared upstream trigger.
P1	Patch boot ordering so time sync completes before OTA scheduler and CAS time validation run; suppress maintenance calculations while clock is epoch/invalid.	Middleware / Irdeto	Removes the plausible self-perpetuating 1970 reboot loop.
P1	Field-test top chronic units: C/N, BER, LNB, cabling, splitter, and 36E lock quality; start with 4664264050, 4664263996, and 4664263958.	Field Ops	Separates external signal path from tuner hardware.
P2	Lab quarantine one high-error box and one explicit-reboot box; swap HDD/tuner board where practical and replay normal OTA/CAS conditions.	Repair Lab	Determines whether chronic units are recoverable or must be replaced.
P2	Add alerting: >3 boot logs per smartcard/day and fleet multi-reboot share >40%; segment chronic vs spike-only cohorts.	Monitoring / Data	Gives early warning before fleet-level customer impact.

Conclusion

Management should treat this as a confirmed service-risk incident, not as isolated customer equipment noise. The evidence supports a two-layer explanation: a shared OTA/CAS/SI/broadcast-side trigger caused the late-May spike, while a subset of decoders with weak signal paths, tuner errors, HDD sensor faults, or boot-time scheduling defects are the most vulnerable. The fastest path to closure is correlation with broadcast/OTA events plus targeted field measurements on the chronic offenders.

Method and Limits

- CSV source: 674 rows covering 98 Explora 3B smartcards from 2026-05-20 to 2026-05-28.

- Raw log source: 125 compressed logs in logs_bundle.zip across 9 smartcards; all were streamed and counted without extracting the ZIP.
- Limitation: the raw ZIP covers chronic units, not all 98 smartcards. It proves mechanisms in the high-risk cohort; it does not alone prove which upstream event triggered every spike-only smartcard.
- Limitation: no headend OTA/CAS/SI change log, weather/uplink record, or field BER/C/N measurement was supplied, so final attribution must remain a ranked RCA pending those checks.